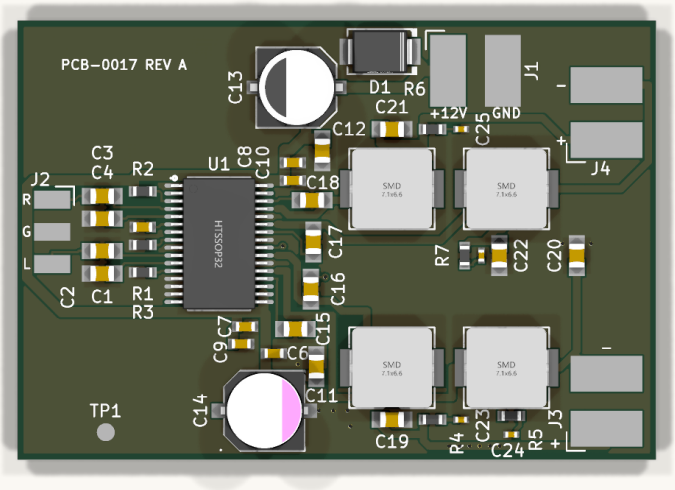
Owner's guide: stereo Class-D amplifier

What the board does, how to wire and power it up, what is on it, what it costs, and what still has to happen before anyone orders it.

PCB-0017-A stereo-class-d-amp is a small, 2-layer stereo Class-D audio amplifier built around the Texas Instruments TPA3118D2, run from a single 12 V supply. It takes stereo line-level audio in on wire pads and drives two bridge-tied speaker outputs. The board reproduces, and is compared against, a published 44 x 34 mm reference design. All four build gates – ERC, placement, routed DRC and DFM – pass. The reference board used the TPA3116D2, sold only in a pad-up package that needs a heatsink bolted on top; this board uses the TPA3118D2 instead, whose pad-down package sinks its heat straight into the board's own copper. That extra ground-pour area is most of why this board, at 53.55 x 36.8 mm, runs about 32% larger than the reference's 44 x 34 mm.



TOP SIDE, AS ROUTED 53.55 X 36.8 MM, 2 LAYERS RENDER

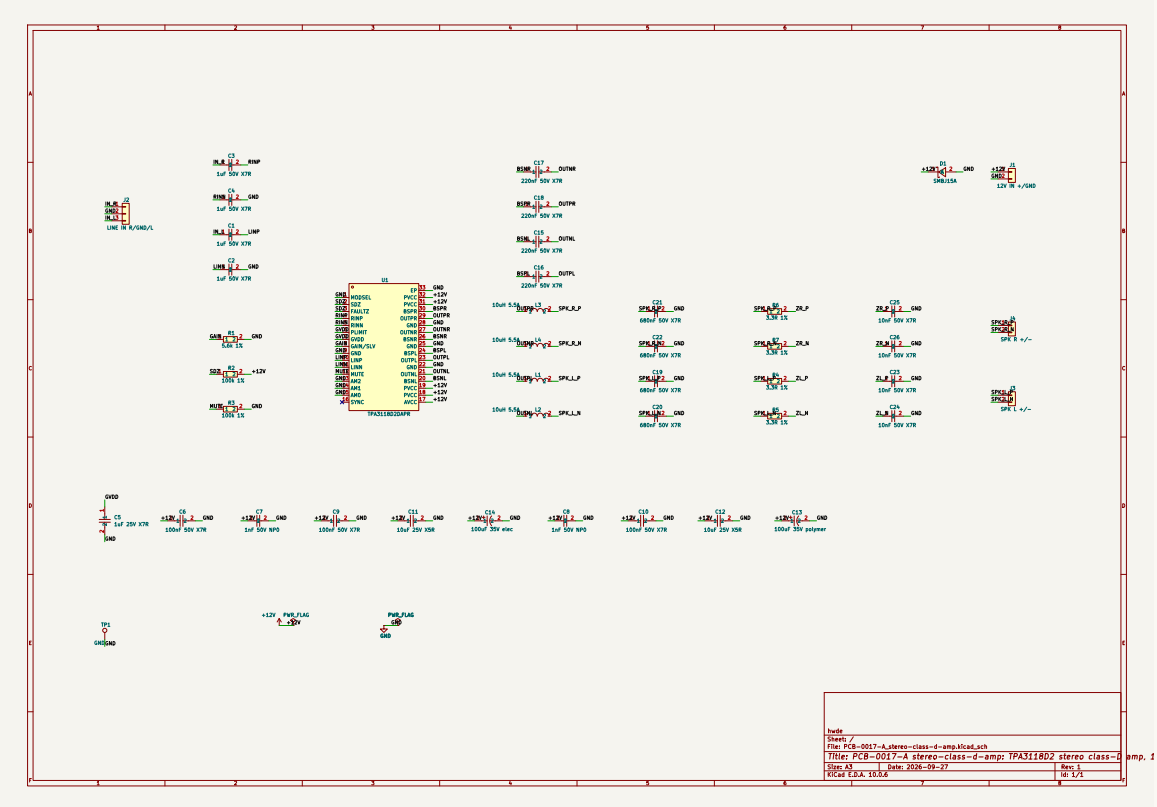
U1 (TPA3118D2) sits centre-board with ground pour on both sides for heat-sinking. J1 (power) and J2 (line in) are on one edge; J3 and J4 (left/right speaker out) and TP1 (GND probe) are on the other.

This is a fab package, not an order: nothing has been sent to JLCPCB. Four proposed verification waivers on U1's power and output pins (see Using it) need the owner's sign-off in `reports/verify-waivers.proposed.json` first; the ordering steps near the end of this guide are written for that later day.

Schematic

One sheet: power in, line in, the amplifier, and the two LC output filters.

FIGURE 1 *stereo-class-d-amp schematic*



Slate wire pads carry power and audio in and out; U1 is the amplifier; the inductor-capacitor pairs on the right are the two output filters.

Power comes in at J1 and splits to two bulk capacitors, C13 and C14, one at each of U1's PVCC pin pairs. Line audio comes in at J2 and is AC-coupled into U1 through C1–C4. U1 drives four switching outputs, one pair per channel, each into its own 10 uH inductor (L1–L4) and a damping branch (R4–R7, C23–C26), which filter the switching carrier before the signal reaches the speaker pads J3 and J4. Every ground on the board is one net, whose bottom-layer pour is both the current return and U1's heat spreader.

Using it

Four wire-pad groups, one probe pad, no switches.

TABLE 1 *connectors*

PAD	CARRIES	NOTES
J1	+12V, GND	10.8–13.2 V DC, sized for 4 A. No reverse-polarity protection: wire it right.
J2	L, GND, R	Stereo line level, about 1 Vrms, AC-coupled.
J3	Left speaker +/-	Bridge-tied load: neither pad is ground.
J4	Right speaker +/-	Bridge-tied load: neither pad is ground.
TP1	GND	Probe pad for a scope or meter ground lead.

Gain is fixed by R1 at 20 dB; there is no volume control, mute switch or status LED, since R2 and R3 hold U1 always on. First power-up: connect a speaker to J3 and/or J4 before applying power, without tying either speaker pad to ground – the output is bridge-tied. Apply 12 V at J1 watching polarity, then feed line audio to J2; TP1 should read J1's GND potential.

Four checks in `reports/verify-waivers.proposed.json` are proposed, not yet approved, on this layout: the 2-layer thermal model under U1 (the basis instead is TI's measured 22 C/W on its 2-layer EVM; with sixteen thermal vias under U1's pad I estimate this board at about 30 C/W, so near 85 C at the 1.5 W design point), two short pin-pitch-limited necks on the +12V and speaker-OUT nets, and one ground-pour check that mis-reads a deliberately split, fully-stitched fill as disconnected. None changes the board; all four wait on the owner's name in the `approved` field of each of that file's eight entries, one per finding.

What's on it

39 fitted parts across 16 BOM lines: 7 unique Basic parts, 9 unique Extended. J1–J4 and TP1 are bare pads, not JLC-assembled parts.

TABLE 2 *bill of materials*

DESIGNATORS	VALUE	LCSC	CLASS	QTY
C1,C2,C3,C4	1uF 50V X7R	C28323	Basic	4
C5	1uF 25V X7R	C106858	Extended	1
C6,C9,C10	100nF 50V X7R	C14663	Basic	3
C7,C8	1nF 50V NP0	C106246	Extended	2
C11,C12	10uF 25V X5R	C15850	Basic	2
C13	100uF 35V polymer	C46550467	Extended	1
C14	100uF 35V elec.	C2992596	Extended	1
C15,C16,C17,C18	220nF 50V X7R	C5378	Basic	4
C19,C20,C21,C22	680nF 50V X7R	C107133	Extended	4
C23,C24,C25,C26	10nF 50V X7R	C15195	Basic	4
D1	SMBJ15A TVS	C113988	Extended	1
L1,L2,L3,L4	10uH 5.5A	C207842	Extended	4
R1	5.6k 1%	C23189	Basic	1
R2,R3	100k 1%	C25803	Basic	2
R4,R5,R6,R7	3.3R 1%	C137725	Extended	4
U1	TPA3118D2DAPR	C46497	Extended	1

Each unique Extended part adds one of JLC's per-part feeder fees on top of the parts themselves (see the assembly line in Table 3). J1, J2, J3, J4 and TP1 are bare copper wire pads with no LCSC part and no connector body; hand wiring in solder, they are not part of JLC's assembly.

What it costs

An estimate and a lower bound, not a quote: get the real number in JLC's own cart before ordering.

Every figure below comes from a reference pricing table, not JLC's live API: measured live prices have run 1.9–3.1x higher than this table. Treat every number here as a floor, and confirm at `https://cart.jlcpcb.com/quote` before ordering.

TABLE 3 *quote matrix, HASL finish*

QTY	PCB	ASSEMBLY	BOARD TOTAL	PER UNIT
5	$2.00	$59.09	$61.09	$12.22
10	$2.09[1]	$60.18[1]	$62.18	$6.22
30	$8.90[1]	$64.53[1]	$73.43	$2.45

[1] Split between PCB and assembly not separately itemised at this quantity in the source data; the board total and per-unit figures are as quoted.

Assembly setup and stencil fees ($16 flat) and per-part feeder fees dominate the cost at low quantity; the bare board itself is a couple of dollars. The parts alone (not counting the PCB or assembly line) are estimated at $2.53 per board. ENIG finish prices the same as HASL in this table at every quantity shown.

Ordering at JLCPCB

Written for after the owner approves the four waivers above – nothing here has been done yet.

Before uploading anything, buy the Extended parts into JLC's inventory for 5 boards: its BOM review can otherwise show one as idle stock, unselected at qty 0, until it has been bought in and the review re-run.

TABLE 4 *pre-buy list, 5 boards*

LCSC	PART	DESIGNATORS	PER BD	BUY
C106858	1uF 25V X7R	C5	1	5
C106246	1nF 50V NP0	C7,C8	2	10
C46550467	100uF 35V polymer	C13	1	5
C2992596	100uF 35V elec.	C14	1	5
C107133	680nF 50V X7R	C19–C22	4	20
C113988	SMBJ15A	D1	1	5
C207842	10uH 5.5A	L1–L4	4	20
C137725	3.3R 1%	R4–R7	4	20
C46497	TPA3118D2DAPR	U1	1	5

Then, in order: (1) upload `PCB-0017-A_stereo-class-d-amp_gerbers.zip`; (2) check the quoted layer count, size and thickness (2 layers, 53.55 x 36.8 mm, 1.6 mm) match the spec above; (3) enable PCB Assembly, top side, for the build quantity; (4) upload `BOM.csv` and `CPL.csv`; (5) confirm every part matched its LCSC number, especially the nine Extended ones just bought in; (6) in the placement preview, check polarity and rotation – no automatic rotation corrections were flagged for this board, but check by eye anyway: D1 (TVS polarity), U1 pin 1, and the C13/C14 polarised capacitors; (7) run JLCDFM as a second opinion; (8) pay. Payment is the owner's step, not a session's.

What it cost to design

Model spend on generating this board – not part of the board's own price.

TABLE 5 *generation cost by step*

STEP	USD
Before the workspace (brief and skill read)	$0.50
Brief and intake	$1.16
Research	$3.84
Architecture	$14.22
Schematic	$3.10
Placement	$4.88
Routing	$0.43
Verification	$0.51
DFM, order files and guide	$0.11
Not in the transcript (side calls)	$68.83
TOTAL	$97.57

The split follows hwde's own record of when each step ran, across ten session transcripts. One session (`811c9ae8`) was still open when this was recorded, so its $0.34 above is its priced-token estimate, not a final figure. About 29.1 million tokens ran on a model (`claude-fable-5-1`) with no listed price; that gap between the step-by-step transcripts and the session's own recorded total is folded into the "side calls" row rather than split further.